
Part 1: HTML Language



Overview

Introduction

- ▶ **HTML = HyperText Markup Language**
 - ▶ Hypertext: text document displayed with references (hyperlinks) to other documents
 - ▶ Markup Language: a computer language using tags to define elements within a document
 - ▶ Language of the web
- ▶ **History**
 - ▶ 1989: First version proposed by Tim Berners-Lee along with WWW
 - ▶ 1995: HTML 2
 - ▶ 1997: HTML 3, **HTML 4**
 - ▶ 2014: **HTML 5**

General Notes

- ▶ Plain text files, with extension .htm or .html
- ▶ HTML is case-insensitive, meaning that either upper or lower case may be used

Working Environment Setup

Working Environment

- ▶ Working with browsers:
 - ▶ Chrome
- ▶ Working with editors:
 - ▶ Notepad
 - ▶ Notepad++
 - ▶ **VS Code**

HTML Basics

First-Page.html

```
<!doctype html>
```

```
<html>
```

```
<head>
```

```
<title>My First HTML Page</title>
```

```
</head>
```

opening tag

closing tag

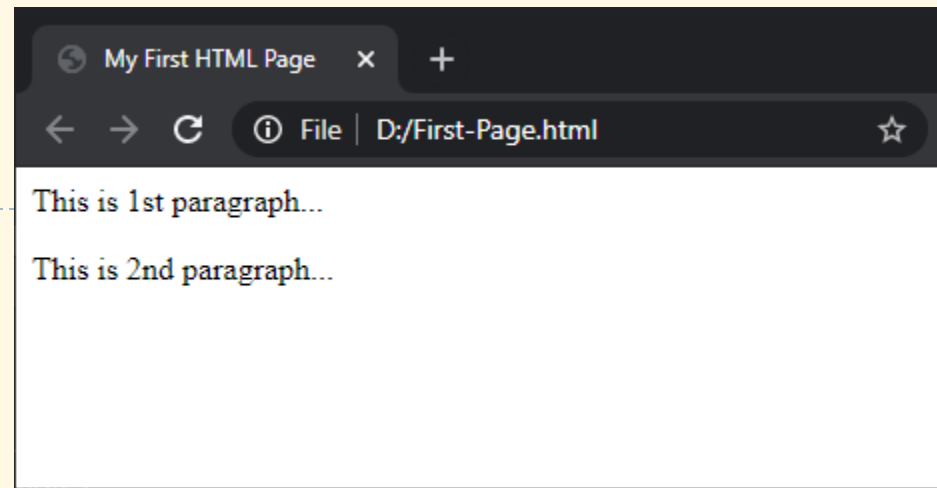
```
<body>
```

```
<p>This is 1st paragraph...</p>
```

```
<p>This is 2nd paragraph...</p>
```

```
</body>
```

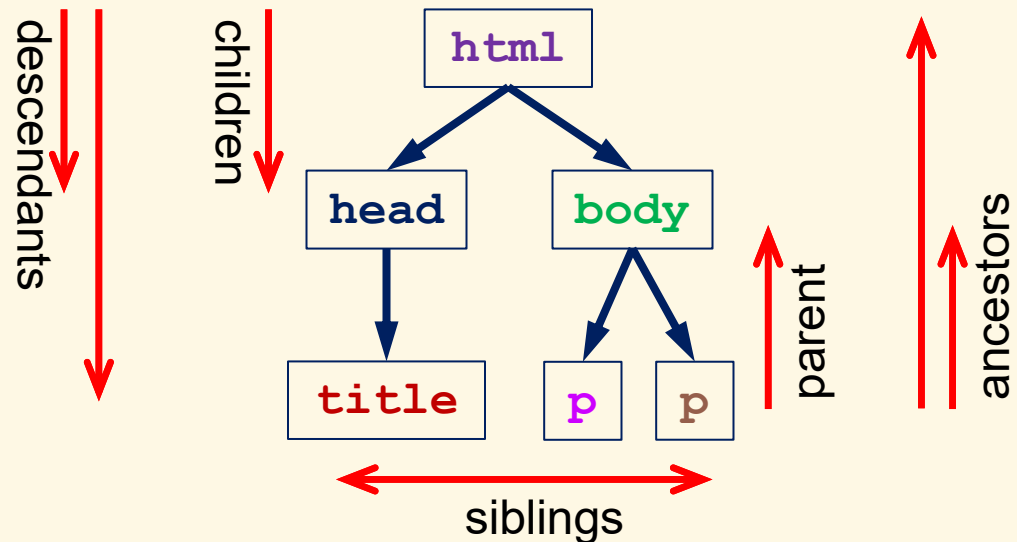
```
</html>
```



Hierarchy of Elements

```
<!doctype html>
<html>
  <head>
    <title>My First HTML Page</title>
  </head>

  <body>
    <p>This is 1st paragraph...</p>
    <p>This is 2nd paragraph...</p>
  </body>
</html>
```



Formatting Tags

<p>This text is emphasized</p>

<p>This text is strong</p>

<p>This text is both

strong and emphasized

</p>

<p>This paragraph has a

new line</p>

self-closing tag

This text is *emphasized*

This text is **strong**

This text is both ***strong and emphasized***

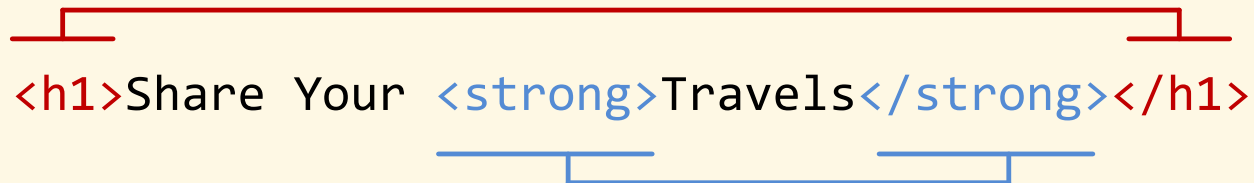
This paragraph has a
new line

- Try some other tags: , <i>, <u>, <small>, <big>, <sub>, <sup>, , <ins>, <mark>, <pre>

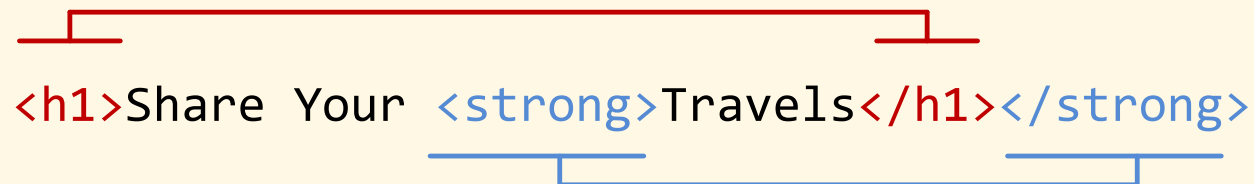
Element Nesting

- ▶ To properly construct a hierarchy of elements, the browser expects each element to be properly nested: a child's ending tag must occur before its parent's ending tag

Correct Nesting



```
<h1>Share Your <strong>Travels</strong></h1>
```

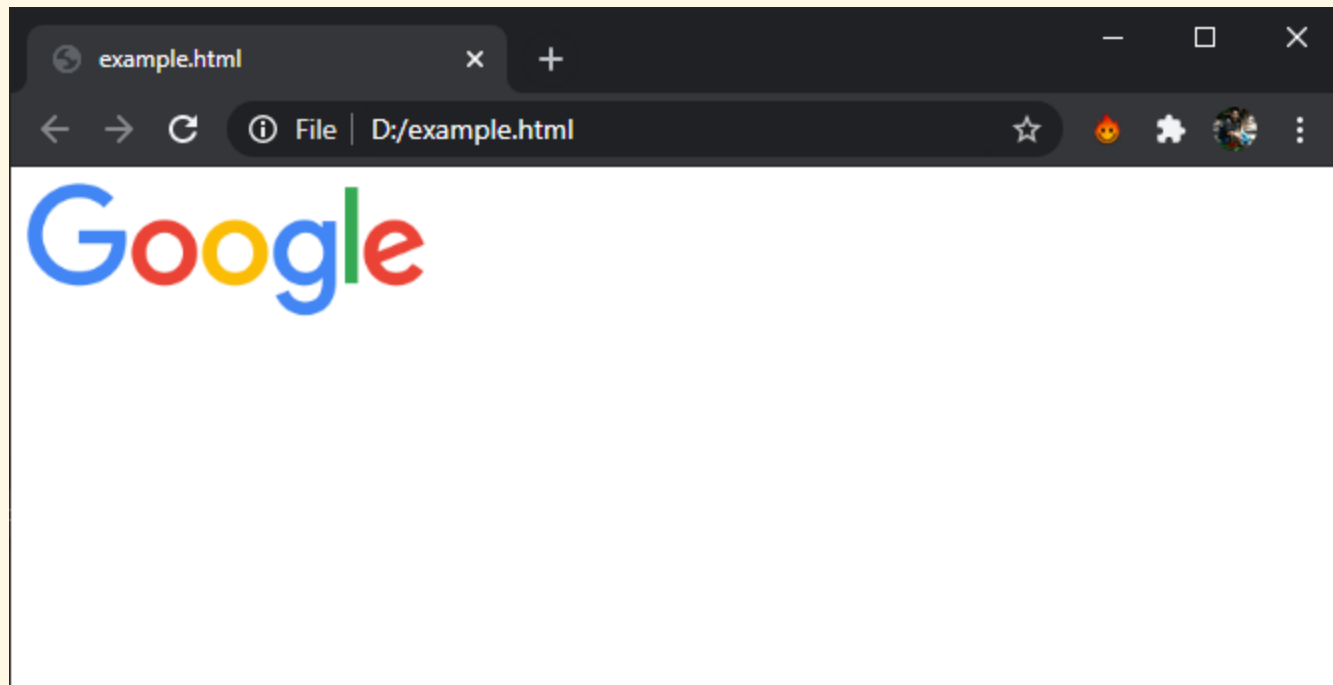


```
<h1>Share Your <strong>Travels</h1></strong>
```

Incorrect Nesting

Images

▶ ``



Elements and Attributes



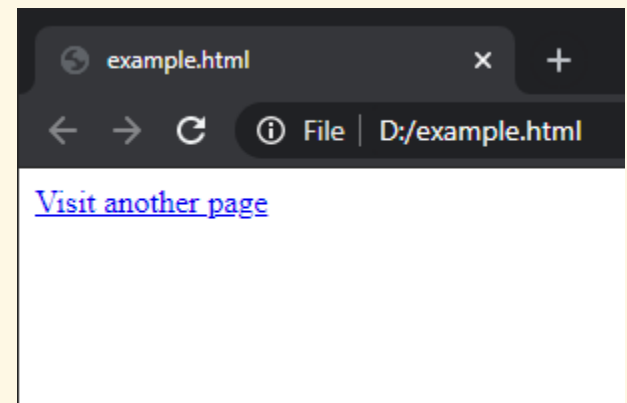
Hyperlinks

- ▶ Links are created using the `<a>` element (a stands for anchor)
- ▶ A link has two main parts: the destination and the label
- ▶ Examples:

- ▶ `<a href="#section2">`
Section 2
`</a>`

- ▶ `<a href="target-page.html">`
Visit another page
`</a>`

- ▶ `<a href="https://www.google.com/fit/">`
Google Fit
`</a>`



Types of Links

- ▶ You can use the anchor element to create a wide range of links:
 - ▶ Links to external sites (or to individual resources such as images or movies on an external site)
 - ▶ Links to other pages or resources within the current site
 - ▶ Links to downloadable resources
 - ▶ Links to other places within the current page
 - ▶ Links to particular locations on another page
 - ▶ Links that are instructions to the browser to start the user's email program
 - ▶ Links that are instructions to the browser to execute a JavaScript function

More Examples of Links

Link to external site

`<a href="http://www.centralpark.com">Central Park</a>`

Link to resource on external site

`<a href="http://www.centralpark.com/logo.gif">Central Park</a>`

Link to another page on same site as this page

`<a href="index.html">Home</a>`

Link to another place on the same page

`<a href="#top">Go to Top of Document</a>`

Link to specific place on another page

`<a href="productX.html#reviews">Reviews for product X</a>`

Link to email

`<a href="mailto://person@somewhere.com">Someone</a>`

Link to javascript function

`<a href="javascript://OpenAnnoyingPopup();">See This</a>`

Link to telephone (automatically dials the number
when user clicks on it using a smartphone browser)

`<a href="tel:+18009220579">Call toll free (800) 922-0579</a>`

URL Absolute Referencing

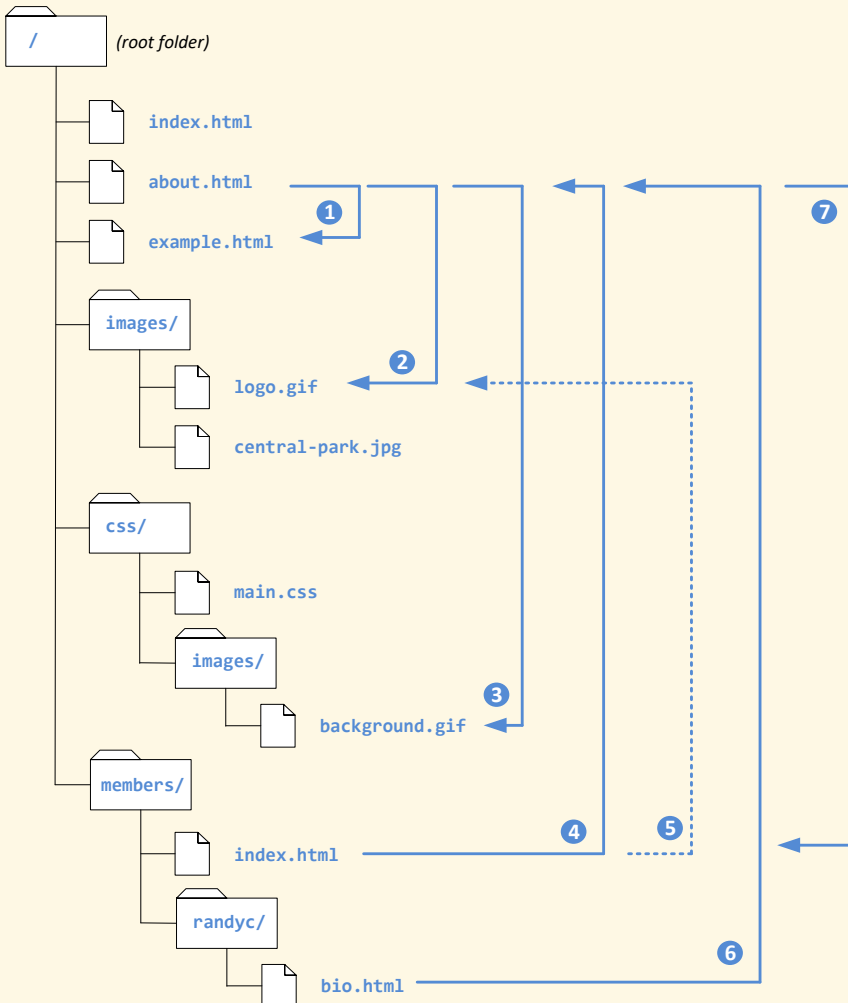
- ▶ When referencing a page or resource on an external site, a full absolute reference is required:
 - ▶ the protocol (typically, `http://` or `https://`)
 - ▶ the domain name
 - ▶ any paths
 - ▶ the file name of the desired resource

URL Relative Referencing

- ▶ We also need to be able to successfully reference files within our site
- ▶ When referencing a resource that is on the same server as your HTML document, then you can use briefer relative referencing. If the URL does not include the “http://” or “https://” then the browser will request the current server for the file
- ▶ The **pathname** tells the browser where to locate the file on the server
 - ▶ Forward slashes (“/”) are used to separate directory names from each other and from file names
 - ▶ Double-periods (“..”) are used to reference a directory “above” the current one in the directory tree

URL Relative Referencing

Share-Your-Travels



Relative Link Type

Example

1 Same Directory

To link to a file within the same folder, simply use the file name.

To link to [example.html](#) from [about.html](#) (in Figure 2.18), use:

```
<a href="example.html">
```

2 Child Directory

To link to a file within a subdirectory, use the name of the subdirectory and a slash before the file name.

To link to [logo.gif](#) from [about.html](#), use:

```
<a href="images/logo.gif">
```

3 Grandchild/Descendant Directory

To link to a file that is multiple subdirectories *below* the current one, construct the full path by including each subdirectory name (separated by slashes) before the file name.

To link to [background.gif](#) from [about.html](#), use:

```
<a href="css/images/background.gif">
```

4 Parent/Ancessor Directory

Use `../` to reference a folder *above* the current one. If trying to reference a file several levels above the current one, simply string together multiple `../`.

To link to [about.html](#) from [index.html](#) in [members](#), use:

```
<a href="../about.html">
```

To link to [about.html](#) from [bio.html](#), use:

```
<a href="../../about.html">
```

URL Relative Referencing

5 Sibling Directory

Use `../` to move up to the appropriate level, and then use the same technique as for child or grandchild directories.

To link to [logo.gif](#) from [index.html](#) in [members](#), use:

```
<a href="../../images/about.html">
```

To link to [background.gif](#) from [bio.html](#), use:

```
<a href="../../css/images/background.gif">
```

6 Root Reference

Root Reference

An alternative approach for ancestor and sibling references is to use the so-called **root reference** approach. In this approach, begin the reference with the root reference (the `/`) and then use the same technique as for child or grandchild directories. **Note that these will only work on the server!** That is, they will not work when you test it

To link to [about.html](#) from [bio.html](#), use:

```
<a href="/about.html">
```

To link to [background.gif](#) from [bio.html](#), use:

```
<a href="/images/background.gif">
```

7 out on your local machine.

Default Document

Web servers allow references to directory names without file names. In such a case, the web server will serve the default document, which is usually a file called [index.html](#) (apache) or [default.html](#)

To link to [index.html](#) in [members](#) from [about.html](#), use either:

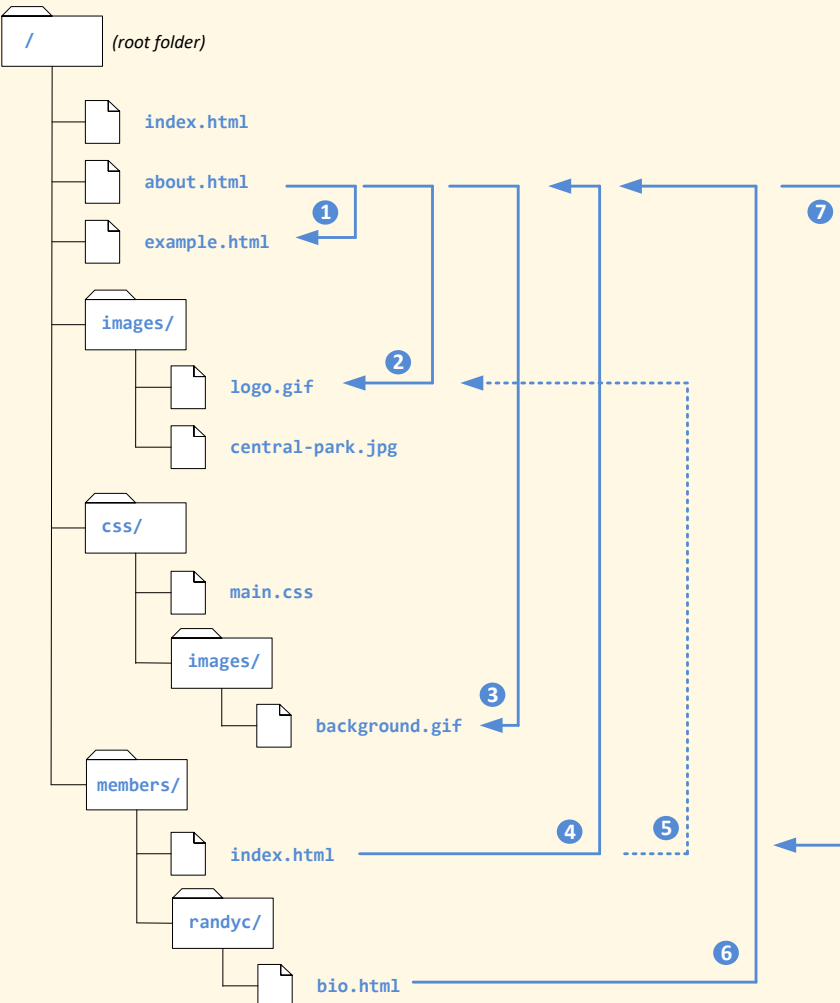
```
<a href="members">
```

Or

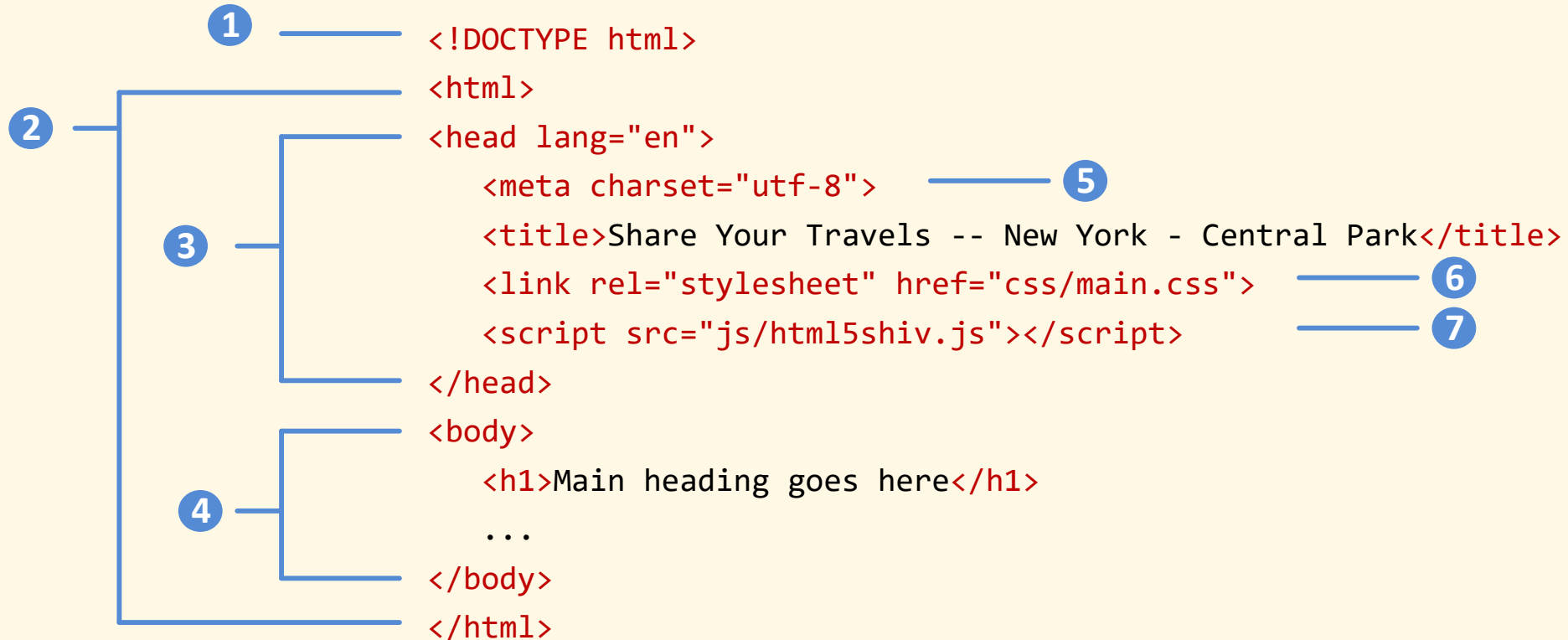
```
<a href="/members">
```

(IIS). Again, this will only generally work on the web server.

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A More Complete Document



!doctype

- ▶ Tells the browser (or any other client software) what type of document it is about to process
 - ▶ Because the browser is capable to process other types: svg, xml,...
- ▶ Notice that it does not indicate what version of HTML is contained within the document: it only specifies that it contains HTML

<html>, <head>, <body>

- ▶ HTML5 does not require the use of the <html>, <head>, and <body>
- ▶ However, in XHTML they were required, and most web authors continue to use them
 - ▶ It is also a good practice to use them
- ▶ The <html> element is sometimes called the root element as it contains all the other HTML elements in the document
- ▶ The <head> contains descriptive elements about the document
- ▶ The <body> contains content that will be displayed by the browser

Inside <head>

- ▶ Notice that the <head> element contains a variety of additional elements
 - ▶ The <meta> element in the example declares that the character encoding for the document is UTF-8
 - ▶ The <link> specifies an external CSS stylesheet file that is used with this document
➔ to be studied in Part 2
 - ▶ The <script> references an external JavaScript file
➔ to be studied in Part 3

Another Example...

<body>

1 `<h1>Share Your Travels</h1>`

2 `<h2>New York - Central Park</h2>`

`<p>Photo by Randy Connolly</p>`

`<p>This photo of Conservatory Pond in`

`<a href="http://www.centralpark.com/">Central Park</a>` 3

`New York City was taken on October 22, 2011 with a`

`<strong>Canon EOS 30D</strong> camera.`

`</p>`

5 `` 4

`<h3>Reviews</h3>`

6 `<div>`

`<p>By Ricardo on <time>September 15, 2012</time></p>`

`<p>Easy on the HDR buddy.</p>`

`</div>`

`<div>`

`<p>By Susan on <time>October 1, 2012</time></p>`

`<p>I love Central Park.</p>`

`</div>`

`<p><small>Copyright © 2012 Share Your Travels</small></p>`

`</body>`

Share Your Travels

New York - Central Park

Photo by Randy Connolly

This photo of Conservatory Pond in [Central Park](http://www.centralpark.com/) New York City was taken on October 22, 2011 with a **Canon EOS 30D** camera.



Reviews

By Ricardo on September 15, 2012

Easy on the HDR buddy.

By Susan on October 1, 2012

I love Central Park.

Copyright © 2012 Share Your Travels

Paragraphs: <p>

- ▶ Paragraphs are the most basic unit of text in an HTML document
- ▶ Notice that the <p> tag is a container and can contain HTML and other inline HTML elements
- ▶ Inline HTML elements refers to HTML elements that do not cause a paragraph break but are part of the regular “flow” of the text

Divisions: `<div>`

- ▶ This `<div>` tag is also a container element and is used to create a logical grouping of content
- ▶ The `<div>` element has no intrinsic presentation
- ▶ It is frequently used in contemporary CSS-based layouts to mark out sections

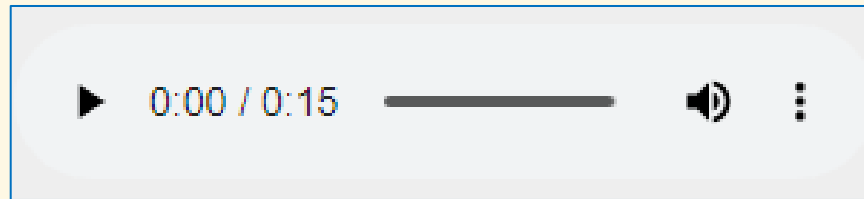
More HTML Elements

Audios: <audio>

<audio controls>

```
<source src="Kalimba.mp3" type="audio/mpeg" />
```

</audio>



- ▶ **More attributes:** autoplay, loop, muted, preload, src

Videos: <video>

```
<video width="500" height="300" controls>  
  <source src="compile.mp4" type="video/mp4" />  
</video>
```

- ▶ **More attributes:** autoplay, loop, muted, preload, src, poster

Inline Frames: <iframe>

- ▶ Used to display external objects including other webpages within a webpage
- ▶ The content inside an iframe exists entirely independent from the surrounding elements
- ▶ Ex:
 - ▶ `<iframe src="hello.html" width="400" height="200"></iframe>`

Block and Inline Elements

- ▶ Block elements add a line break before and after them
 - ▶ Ex: `<div>`, `<table>`, `<hr>`, headings, lists, `<p>`...
- ▶ Inline elements don't break the text before and after them
 - ▶ Useful for modifying a specific portion of text
 - ▶ Ex: `<span>`, `<a>`, `<em>`, `<strong>`, `<b>`, `<img>`, `<audio>`, `<video>`, `<iframe>`...

Character Entities

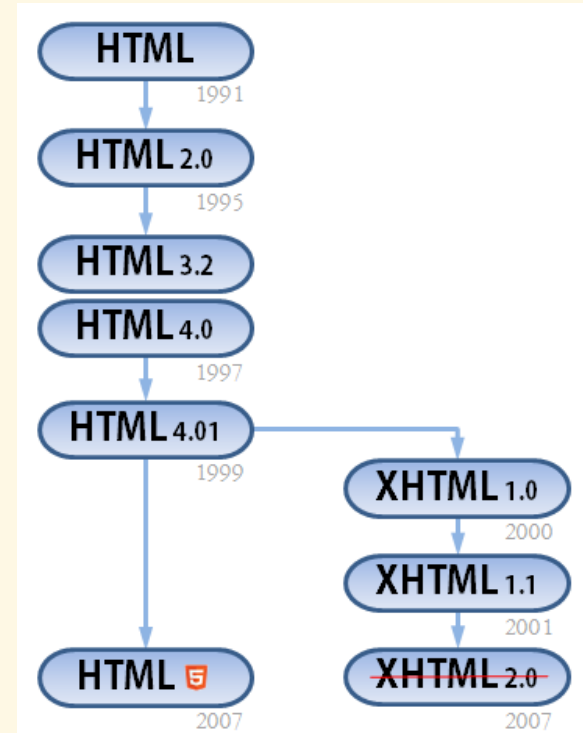
- ▶ Special characters for symbols for which there is either no way easy way to type in via a keyboard (such as the copyright symbol or accented characters) or which have a reserved meaning in HTML (for instance the “<” or “>” symbols)
- ▶ They can be used in an HTML document by using the entity name or the entity number
- ▶ Ex:
 - ▶ ` `
 - ▶ `©`
 - ▶ `>`
 - ▶ `<`
 - ▶ `ୄ`

Comments

- ▶ Pieces of code which are ignored by any web browser
- ▶ It is a good practice to add comments, to indicate sections of a document, and any other notes to anyone looking at the code
- ▶ HTML comments are placed in between `<!-- ... -->` tags
- ▶ Ex:
 - ▶ `<!-- Comment text`
can span through as many as
lines you like `-->`

XHTML vs. HTML

- ▶ XHTML is stricter than HTML
 - ▶ Tags and attribute names must be in lowercase
 - ▶ All tags must be closed (`<br/>`, `<img/>`) while HTML allows `<br>`, `<img>` and implies missing closing tags (`<p>par1 <p>par2`)
 - ▶ XHTML allows only one root `<html>` element (HTML allows more than one)
- ▶ Many element attributes are deprecated in XHTML, most are moved to CSS
- ▶ Attribute minimization is forbidden, e.g.
 - ▶ `<input type="checkbox" checked>`
 - ▶ `<input type="checkbox" checked="checked" />`
- ▶ XHTML is too restricted and unforgiving for minor errors → retired by the W3C in 2018



Headings

Headings

- ▶ HTML provides 6 levels of heading (<h1>, <h2>, <h3>, ..., <h6>), with the higher heading number indicating a heading of less importance
- ▶ Headings are an essential way for document authors use to show their readers the structure of the document
- ▶ Good practice:
 - ▶ Don't use headings to make your text look BIG or **bold**; use them only for highlighting the heading of your document and to show the document structure
 - ▶ Search engines use headings to index the structure and content of the web pages so use them very wisely in your webpage

Example

`<h1>Heading level 1</h1>`

`<h2>Heading level 2</h2>`

`<h3>Heading level 3</h3>`

`<h4>Heading level 4</h4>`

`<h5>Heading level 5</h5>`

`<h6>Heading level 6</h6>`

Heading level 1

Heading level 2

Heading level 3

Heading level 4

Heading level 5

Heading level 6

Lists

Lists

- ▶ **Unordered lists:** Collections of items in no particular order; these are by default rendered by the browser as a bulleted list
- ▶ **Ordered lists:** Collections of items that have a set order; these are by default rendered by the browser as a numbered list
- ▶ **Definition lists:** Collection of name and definition pairs (Ex: FAQ list)

Unordered Lists

```
<ul>
  <li>Chocolate Cake</li>
  <li>Black Forest Cake</li>
  <li>Pineapple Cake</li>
</ul>
```

- Chocolate Cake
- Black Forest Cake
- Pineapple Cake

circle, disc, square, none

```
<ul style="list-style-type:square">
  <li>Chocolate Cake</li>
  <li>Black Forest Cake</li>
  <li>Pineapple Cake</li>
</ul>
```

- Chocolate Cake
- Black Forest Cake
- Pineapple Cake

Ordered Lists

```
<ol>
  <li>Fasten your seatbelt</li>
  <li>Starts the car's engine</li>
  <li>Look around and go</li>
</ol>
```

1. Fasten your seatbelt
2. Starts the car's engine
3. Look around and go

**lower-alpha, lower-greek, lower-latin, lower-roman,
upper-alpha, upper-greek, upper-latin, upper-roman,
decimal**

```
<ol start="10" style="list-style-type:upper-roman">
  <li>Fasten your seatbelt</li>
  <li>Starts the car's engine</li>
  <li>Look around and go</li>
</ol>
```

- X. Fasten your seatbelt
- XI. Starts the car's engine
- XII. Look around and go

Definition Lists

```
<dl>
```

```
  <dt>Bread</dt>
```

```
  <dd>A baked food made of flour.</dd>
```

```
  <dt>Coffee</dt>
```

```
  <dd>A drink made from roasted coffee beans.</dd>
```

```
</dl>
```

Bread

A baked food made of flour.

Coffee

A drink made from roasted coffee beans.

Exercise

- Create HTML pages presenting the following contents:

About Victoria Kirst

My name is Victoria and I am jolly, clumsy, and four-eyed.

My Classes This Quarter

- CSE 451 - Operating Systems
- CSE 471 - Computer Design and Organization
- PHYS 121 - Physics: Mechanics
- CSE 498 - Research w/ Prof. Luis Ceze

My Favorite Movies

(I actually don't watch too many movies, so...here goes!)

1. The last 30 minutes of Forrest Gump ([IMDB](#))
2. Star Trek Episode V with Zazu ([IMDB](#))
3. Fight Club (not really, but I've seen like 3 movies total so this is my 3rd fave by technicality) ([IMDB](#))

My Moods



Happy:



Sad:

Fun Facts About My Neighbors

- Sue Smith: Effervescent is a word that describes her.
- Bill Thompson: Loves playing Yu-Gi-Oh.



Meet Jack



This is **Jack**. He's an odd little boy that just got an empty box for his birthday. He may *look* happy, but he's a *little* disappointed. Now he'll have to put a few things in the box.

Jack would really like...

- Roller blades
- Magic tricks
- A motorized Lego set

You can get all this (and more) at an online toy store. Two very popular ones are toysrus.com and [eToys](http://eToys.com).

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Tables

Definitions

- ▶ A table is a matrix formed by the intersection of horizontal rows and vertical columns

	Column 1	Column 2	Column 3	Column 4
Row 1				
Row 2		cell		
Row 3				

- ▶ The intersection of a column and row is called a **cell**

<table>, <tr>, <th>, <td>

- ▶ HTML expects everything between <table> and </table> to be part of the table
- ▶ <tr>: a **row container**
 - ▶ Within each row container, each cell is defined by either table header <th>, or table data <td>
 - ▶ <th> is for **table header**, having exactly the same output as the combination: <td><center>
 - ▶ <td> is for ordinary **table data**

colspan, rowspan Attributes

- ▶ `colspan="x"`: forces the cell to cover x columns (default = 1)
- ▶ Ex:
 - ▶ `<td colspan="3"> . . . </td>`
- ▶ `rowspan="y"`: forces the cell to cover y rows (default = 1)
- ▶ Ex:
 - ▶ `<td rowspan="2"> . . . </td>`

Other Table Options

- ▶ `<thead>` defines a header section, contains `<tr>` elements
 - ▶ If used, it MUST precede a `<tbody>`
- ▶ `<tbody>` contains the usual table rows, and this should be followed by a footer
- ▶ `<tfoot>` defines a footer section
- ▶ These three are optional unless the first is used

Example

```
<table>
  <thead>
    <tr> <th colspan="3" align="center">Header</th> </tr>
  </thead>

  <tbody>
    <tr> <th>Row Header</th> <td>Data</td> <td>Data</td> </tr>
    <tr> <td>Data</td> <td>Data</td> <td>Data</td> </tr>
  </tbody>

  <tfoot>
    <tr> <th colspan="3" align="center">Footer</th> </tr>
  </tfoot>
</table>
```

Exercise

- Create a HTML presenting the following table:

		Clothes			Accessories	
		Trousers	Skirts	Dresses	Bracelets	Rings
Belgium	Antwerp	56	22	43	72	23
	Gent	46	18	50	61	15
	Brussels	51	27	38	69	28
The Netherlands	Amsterdam	89	34	69	85	38
	Utrecht	80	12	43	36	19

All HTML Elements

- ▶ Reference page:

- ▶ <https://developer.mozilla.org/en-US/docs/Web/HTML/Element>

Styles

Inline Styles

- ▶ It is possible to change the appearance and some behaviors of elements using `style` attribute:
 - ▶ `<p style="background-color:red">...</p>`
 - ▶ Multiple style properties can be combined
 - ▶ `<p style="color:blue; font-size:2em">...</p>`
 - ▶ Common style properties:
 - ▶ `color, background-color, font-family, font-size, font-style, font-weight`
 - ▶ `text-align, vertical-align, display, float`
 - ▶ `height, width, border, margin, padding`
- ➔ To be studied in Part 2